Chapter 3 Atomic Physics

Importance of Hydrogen Atom

- Simplest Atom
- Can be used to describe the allowed states of more complex atoms, which enables us to understand the periodic table.
- Ideal system for performing precise comparisons of theory with experiment.
- Can be extended to other single-electron ions.$(\mathrm{Li}^{2+}, \mathrm{He}^{+})$

Base of Atom Structure

J.J Thomson

A volume of positive charge and electrons embedded throughout the volume.

Rutherford

Through scattering experiments, a few the alpha particles deflected from the original paths or even reversed.

Rutherford's Model

- Planetary model
- Based on results of foil experiments.
- Positive charge is concentrated in the center of the atom, called the nucleus.
- Electrons orbits the nucleus like planets orbit the sun.

Difficulties with the model

- Can not explain the discrete characteristic frequencies of electromagnetic radiation.
- Rutherford's electrons are undergoing a centripetal acceleration so should radiate electromagnetic waves of the same frequency. And the radius should steadily decrease as the radiation is given off and eventually the electron spiral into the nucleus but it doesn't.

Emission Spectra

A gas has a voltage at a low pressure, which makes the gas emits light characteristic of the gas. When the light is analyzed with a spectrometer, a series of discrete bright line is observed.

Emission Spectrum of Hydrogen

The wavelengths of hydrogen's spectral lines can be found from

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

where R_H is a constant, n is a positive integer. Balmer Series has lines given by this equation.

Absorption Spectra

A element can also absorb light at specific wavelengths. Absorption Spectra can be obtained by passing a continuous radiation spectrum through a vapor of the gas. And a series of dark lines superimposed on the spectrum, which coincide with the bright lines of the emission spectrum.

Neils Bohr

The Bohr Theory of Hydrogen

His model includes both classical and non-classical ideas and want to explain why the atom was stable.

The electron orbits around the proton(质子) under the influence of Coulomb(库仑) force, which produces the centripetal acceleration.

Only certain electron are stable, which do not emit energy in the form of electromagnetic radiation and confirm classical mechanics.

Radiation is emitted when electron jumps from a more energetic initial state to a lower state. (Cannot be treated classically)

The frequency emitted by the jump is related to the change in the atom's energy, which is not the same as the frequency of the electron's orbital motion(classically). And it is given by

$$E_i - E_f = hf$$

Mathematics of Bohr's Assumptions and Results

Electron's orbital angular momentum:

$$rmv = n\hbar, L = n\hbar, \text{ for } n = 1, 2, 3, \cdots; \hbar = h/2\pi$$

where n is main quantum number. And the size of electron orbits is determined by a condition imposed on the electron's orbital angular momentum.

Bohr Radius

The radius of the Bohr orbits are:

$$r = \frac{h^2\epsilon_0}{\pi m e^2} n^2, \text{for } n = 1, 2, 3, \cdots$$

for abbreviation,

$$r = a_0 n^2$$

where

$$a_0 = \frac{h^2\epsilon_0}{\pi m e^2} \approx 52.92\text{pm}$$

And this explains the electron can only exist in certain allowed orbits determined by the integer n.

And when $n = 1$, the orbit has the smallest radius called the Bohr radius a_0.

The total energy of the atom:

$$E = K + U = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

and by the centripetal force and angular momentum the energy of any orbits is

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2} \frac{1}{n^2}, \text{ for } n = 1, 2, \cdots = -\frac{13.6\text{eV}}{n^2}$$

Specific Energy Levels

The lowest energy state is called the ground state, corresponding to $n = 1, E = -13.6\text{eV}$. And when $n \geq 2$, it is called excited state.

The ionization energy is the energy needed to completely remove the electron from the atom. (13.6 eV for hydrogen) And the uppermost level corresponds to $E = 0$ and $n \to \infty$.

Generalized Spectral Equation

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

for Balmer series, $n_f = 2$; for Lyman series, $n_f = 1$.

When an transition occurs between two state, photon(光子) is emitted and its frequency is

$$f = \frac{E_i - E_f}{h}$$

Bohr's Correspondence Principle

Quantum mechanics is in agreement with classical physics when the energy differences between quantized levels are very small.

Successes of the Bohr Theory

- Explained several features of the hydrogen spectrum
  - accounts for Balmer and other series
  - predicts a value for R_H that agrees with experimental value
  - gives an expression for the radius of the atom
  - predicts energy levels of hydrogen
  - gives a model of what the atom looks like and how it behaves.
- Can be extended to "hydrogen-like" atoms(with one electron)

Limitations of the Bohr Theory--Fine Structure

Spectral lines are in fact two or more closely spaced lines without magnetic field. This splitting is called fine structure.

The spin magnetic quantum number, m_s, is introduced to explain the fine structure.($\pm\frac{1}{2}$)

Sommerfeld

In order to explain the fine structure, Sommerfeld improved Bohr's atomic structure.

- Revise the circle to elliptical orbits
- Add the orbital quantum number l.
- All states with the same principle quantum number n are said to form a shell. And the states with the same principle of n and l are said to form a subshell.

Zeeman Effect

Modification of the Bohr Theory--Zeeman Effect

Zeeman effect is the splitting of spectral lines in a strong magnetic field. It indicates that the energy of an electron is slightly modified when the atom is immersed in a magnetic field.

And the orbital magnetic quantum number, m_l is introduced, varying from $-l$ to $+l$ in integer steps.

De Broglie Waves

The assumption of Bohr's does not explain why the restriction occurred.
de Broglie assume that electron orbit is stable only if it contained an integral number of electron wavelengths. That is

$$2\pi r = n\lambda$$

And it confirms the quantization of angular momentum and explains the appearance of integers in Bohr's equations as a natural consequence of standing wave patterns.

And it is the first convincing argument that the wave nature of matter was the heart of the behavior of atomic systems. Schrodinger's wave equation was subsequently applied to atomic systems.

Schrodinger's Equation

Give the mathematic equations to calculate the wave equation of hydrogen atom.

Quantum Mechanics and the Hydrogen Atom

The quantum mechanics use directly mathematics and not from any assumptions made to make the theory agree with experiments.

n: the principle quantum number, for the energy of the state.
l: the orbital quantum number, for the magnitude of angular momentum.
m_l: the orbital magnetic quantum number, the orientation in space of angular momentum vector.

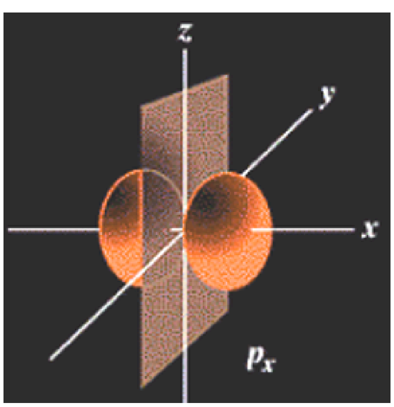
Electron Clouds

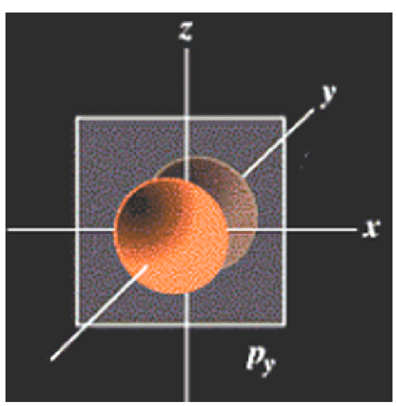
The electron is not confined to a particular orbital distance. And the probability of finding the electron at the Bohr radius is maximum.

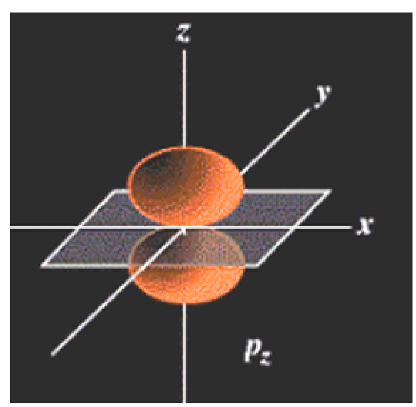
The electron clouds represents the probability of the electron appear at the position. And the densest regions of the cloud represent the highest probability of finding electron.

p orbitals

Look like a dumbell with 3 orientations : p_x, p_y, p_z.

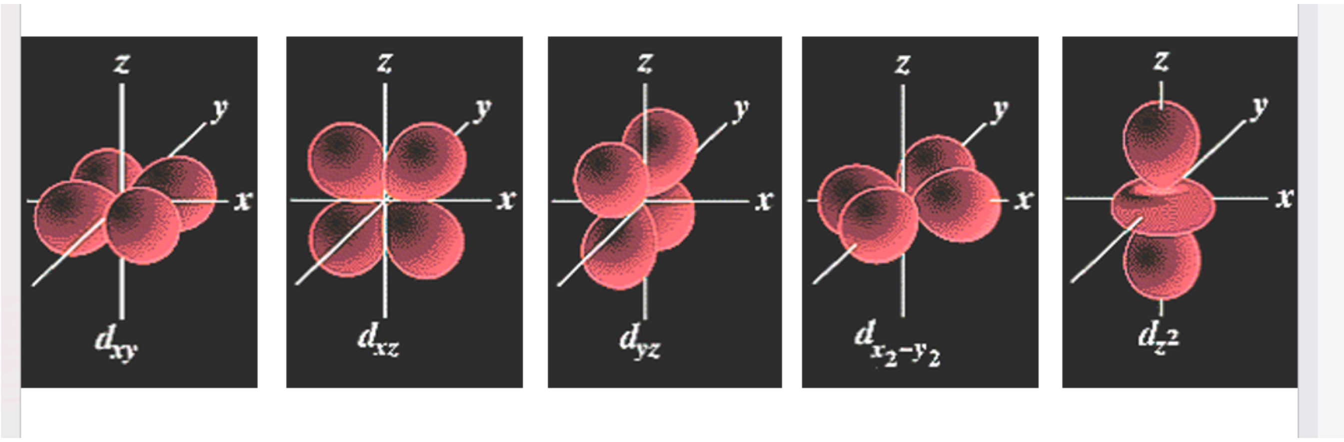






d orbitals

4 look like two dumbells in a clover shape, 1 looks like a dumbells wrapped by a donut.



The Pauli Exclusion Principle

No two electrons in an atom can ever have the same set of the quantum numbers n, l, m_l, m_s.

- Electron goes into the vacant subshell in lowest energy first.
- Otherwise, electron radiate energy until reaching the subshell with the lowest energy.
- A subshell is filled when holding $2(2l+1)$ electrons.
- A shell is filled when holding $2n^2$ electrons.

Spin Magnetic Quantum Number

The electron is not physically spinning. And there are two directions for spin: $m_s = 1/2$ for spinning up, $m_s = -1/2$ for spinning down. And there is slight energy difference between two spins and accounts for the doublet in some lines.

Paul Dirac uses Dirac Equation predicted the existence of the position.

Spin-Orbit Interaction

To explain the split of p-d-f orbitals.

Crystal-field theory

To explain the split of d orbitals.

The Periodic Table

The outermost electrons are primarily responsible for chemical properties of the atom. Mendeleev arrange the elements, based on Pauli's Exclusion principle.

Characteristic X-Rays

When a metal target is bombarded by high-energy electrons, x-rays are emitted(with secondary electron and auger electron).

When a metal target is bombarded by X-ray, photoelectrons and Compton-scattered electrons are emitted.

X-ray Spectrum typically consists of a broad continuous spectrum and a series of sharp lines, depending on the metal. And the lines are called characteristic x-ray.

Explanation of Characteristic X-rays

The bombarding electron collides with an electron in the target metal that is in a inner shell. If energy is sufficient, the electron is removed from the target atom. And the vacancy is filled by an electron from a higher energy level. And the transition emit the photon whose energy is equal to the difference between the two levels.

Moseley Plot and Atomic Number

Moseley ensures the atomic number is the most important factor in determining its chemical and physical properties and modifies some elements.

Atomic Transitions

Energy levels

An atom may have many possible energy levels, most is in the ground state. And only photons with energies corresponding to differences between energy levels can be absorbed.

Stimulated Absorption

When absorbed a photon with ΔE, on electron jumps to a higher energy level(excited states). And $\Delta E = hf = E_2 - E_1$.

Spontaneous Emission

An atom in an excited state has a constant probability to jump back to a lower state and emitting a photon.

Stimulated Emission

An atom is in an excited state and a incoming photon is incident on it, which increases the probability to return the ground state. And two emitted photons, incident one and stimulated one.

Basic Principle of XPS

$$KE = hv - BE - \phi$$

where KE is the kinetic energy of emitted photoelectrons, hv is the energy of incoming photon, BE is the binding energy of electrons, ϕ is work function.(Known)

Population Inversion

More atoms are in higher states rather than lower states, which is called a population inversion.

Population inversion is the prerequisite for laser generation.

Lasers

Three conditions must be met:

- The system must be in a state of population inversion
- The excited state of the system must be a metastable state
- The emitted photons must be confined in the system long enough to allow them to stimulate further emission from other excited atoms.(Achieved by using reflecting mirrors)

Laser Beam of He, Ne

A high voltage is used to produce excited states. And the mixture of helium and neon is confined to a glass tube sealed at the ends by mirrors.

Energy Bands in Solids

In solids, the discrete energy levels of isolated atoms broaden into allowed energy bands separated by forbidden gaps.

The separation and the electron population of the highest bands determine whether the solid is a conductor, insulator or semiconductor.

Energy Level Definitions

- valence band: The highest filled band.
- conduction band: The next higher empty empty band.
- energy gap: The energy difference between the top of the valence band and the bottom of the conduction band.

Conductors

It takes very little energy for electrons to jump from the partially filled to one of the nearby empty states. And it is easy to conduct electrons.(valence band is not filled or valence band is overlap to conduction band, energy gap are very little)

Insulators

A large band gap separates the valence and conduction bands. And the valence band is completely full of electrons. And it is difficult to conduct electrons.

Semiconductors

A small energy gap. And Thermally excited electrons have enough energy to cross the band gap. The resistivity of semiconductors decreases with increases in temperature.

Holes are empty states in the valence band created by electrons that have jumped to the conduction band. And some electrons in the valence band move to fill the holes and therefore also carry current. And the valence electrons that fill the holes leave other holes.

Electrons move toward the positive electrode, holes move toward the negative electrode. (Symmetrical current process)

Doping in Semiconductors

Doping is the adding of impurities to a semiconductor, resulting in both the band structure and the resistivity being changed.

n-type Semiconductors

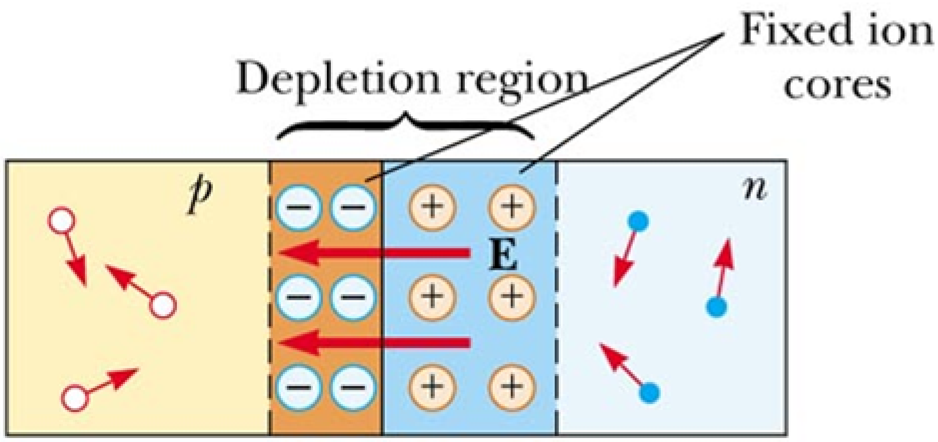
Donor atoms doped in the semiconductors contain one more electron than the semiconductor material. Free electrons are main current carrier and the donor atoms form a positive ion that cannot move freely.

p-type Semiconductors

Acceptor atoms doped in the semiconductors contain one less electron than the semiconductor material. Hole(Equivalent to positive charge) are main current carrier and the acceptor atoms form a negative ion that cannot move freely.

A p-n Junction

It is formed when a p-type semiconductor is joined to an n-type and three regions exits: a p region, an n region and a depletion region.



The Depletion Region

Holes from the p side diffuse to the n side and leave fixed negative ions and electrons from the n side diffuse to the p side and leave fixed positive ions. that results in depletion of mobile charge carriers.

And it has the ability to pass current in only one direction.

Diode Action

When p-side is connected to a positive terminal, the device is forward biased and current flows. When n-side is connected to a positive terminal, the device is reverse biased and very small reverse current results.

Applications of Semiconductor Diodes

- rectifiers
- transistors
- integrated circuit